Part III

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III

10 The Hip

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Abstract
The aim of this chapter is to provide a systematic approach to identifying pathologies of the hip. In this chapter we review hip anatomy in the sagittal, coronal and axial planes and describe the most common pathologic conditions seen in the hip. In addition, this chapter will detail MRI findings seen in a wide array of hip pathologies that physicians must be wary of. By the end of this chapter, the reader will be able to determine the ideal MRI technique required to diagnose a hip pathology and recognize characteristic findings associated with it.

Keywords: hip, magnetic resonance imaging (MRI), hip imaging, femoral-acetabular impingement, intra-articular hip pathology

10.1 Normal Anatomy

When evaluating the hip, it is helpful to divide the normal hip anatomy into four distinct layers: layer 1, the osteochondral layer; layer 2, the capsulolabral layer; layer 3, the muscular layer; and layer 4, the neural layer.[1] Magnetic resonance imaging (MRI) offers the ability to evaluate each of these layers in detail.

10.1.1 Sagittal Anatomy

The sagittal images of the hip allow visualization of the gluteus maximus, medius, and minimus muscles laterally; the short external rotator tendons as they insert on or near the greater trochanter; and as the hip joint comes into view, the cartilage of the acetabular weight-bearing dome, the iliopsoas anterior to the femoral head, and the anterior acetabular labrum. One can often also see the hamstring origin and the sciatic nerve, depending on the extent of the scan. When evaluating the anterior acetabular labrum for evidence of tear or degeneration, the sagittal view will provide the best visualization (▶ Fig. 10.1).



Fig. 10.1 A sagittal fast spin-echo (FSE) image of the hip showing a normal anterior labrum (*left arrow*) as a triangular focus of low signal and normal articular cartilage of the acetabulum (*top arrow*). (Source: © 2009 Hospital for Special Surgery, New York, NY.)

10.1.2 Coronal Anatomy

The coronal images demonstrate the pubic symphysis and the aponeurotic plate of the adductors anteriorly, the cartilage and acetabular labrum of the hip joint, the attachment of the gluteus medius and minimus tendons on the greater trochanter, and the hamstring origins from the ischial tuberosity posteriorly. The ligamentum teres, iliac wing (including abductor and sartorius origins), iliopsoas bursa, and rectus femoris origin can be identified. Coronal imaging gives the best view of the weight-bearing surface over the femoral head, as well as the lateral dome of the acetabulum and the superior labrum (▶ Fig. 10.2).

10.1.3 Axial Anatomy

Axial anatomy of the hip is useful to identify. From superior to inferior, it consists of the psoas and iliacus muscles, the iliopsoas tendon anterior to the femoral head, the acetabular labrum, and the sciatic nerve as it exits the sciatic notch. Axial imaging can demonstrate the pubis, the short external rotators, adductors, and rectus femoris and sartorius muscles and can rule out other causes of pelvic area pain, including obturator and inguinal hernias (▶ Fig. 10.3).

10.2 Specialized Pulse Sequences and Imaging Protocols

As with MRI of any anatomic region, accurate MRI assessment of the hip requires attention to technical detail. Traditionally, MRI of the hip performed for suspicion of osteonecrosis or an occult fracture used large fields of view, which provided poor in-plane resolution and little to no detail of the articular cartilage or labrum. More recently, improvements in MRI technology allow small fields of view and high spatial resolution. Survey images of

the entire pelvis are still helpful for screening for other causes of pain such as the following:

- Muscle strains.
- Pelvic masses.
- Hernias.
- Spine and sacroiliac (SI) joint pathology.
- Occult fractures.

At the authors' institution, intermediate TE (proton-density) sequences are used in each plane with high resolution (512 × 320–384) and a maximum field of view of 20 cm. These are obtained in the sagittal and coronal planes and an oblique axial image through the femoral neck. Sometimes radial images through the femoral neck are added. Survey images of the pelvis using a coronal fat-suppressed sequence (usually inversion recovery) and an axial proton-density sequence are obtained.

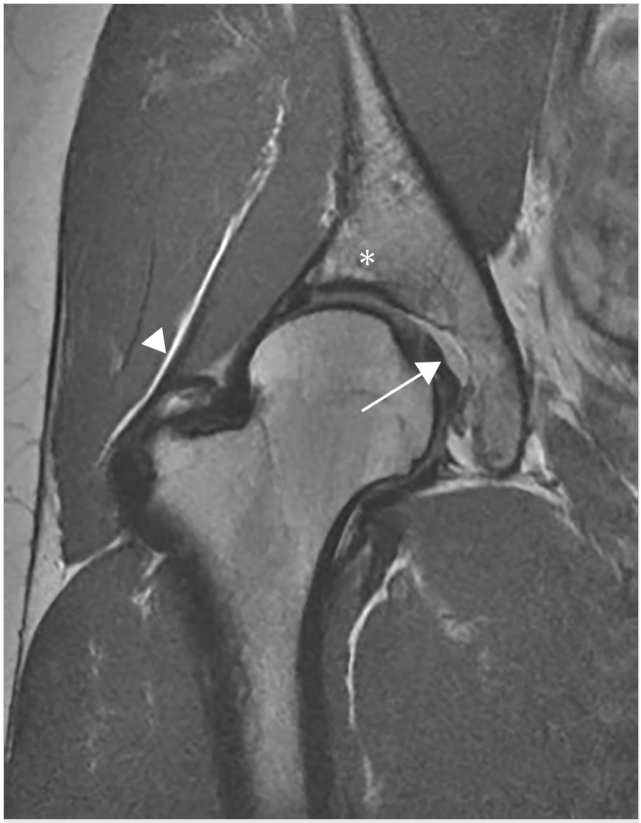


Fig. 10.2 A coronal fast spin-echo (FSE) of the hip showing a normal ligamentum teres (*arrow*), acetabular labrum (*asterisk*), and attachment of the gluteus minimus tendon (*arrow head*).

At the authors' institution, use of the intermediate TE sequence rather than T1-weighted and T2-weighted images is preferred for several reasons. First, T1 is poor for evaluating cartilage because there is no contrast between fluid and cartilage. Second, it is also poor for evaluating subtle muscle abnormalities, so another sequence in each plane would be required for a thorough evaluation, and yet a third to evaluate cartilage. The intermediate TE sequence eliminates this redundancy by allowing enough signal for excellent resolution and enough contrast to evaluate all structures (bone, ligament, tendon, muscle, and articular cartilage), as well as the quality of synovitis, if present. One can obtain this sequence in each plane and then add one fat-suppressed sequence to evaluate for subtle findings and to assess acuity.

Many authors advocate using MR arthrography of the hip for the evaluation of labral pathology and articular cartilage (see chapter 5, Articular Cartilage),[2,3,4] but this modality increases the cost and imaging time and makes MRI an invasive procedure.[5] Although noncontrast imaging has been shown to identify labral and chondral pathology and is becoming more popular as new technology allows detailed evaluation, many institutions in the United States and abroad continue to use MR arthrography to improve the visualization of intra-articular hip pathology, especially labral pathology.[6,7]

10.3 Systematic Approach to Image Review

Beginning with the fat-suppressed images, we look for areas of bone edema, as may be seen with a stress reaction or stress fracture of the femoral neck (▶ Fig. 10.11), osteonecrosis, subchondral fracture, or advanced arthrosis. Then we examine the greater trochanteric region for trochanteric bursitis. The remainder of the evaluation is done using proton-density images, with confirmation using the fat-suppressed images. First we look at the abductor insertions on the coronal images, and then we use both coronal and sagittal images to evaluate the labrum and articular cartilage. A torn labrum appears as a line of high signal interrupting the continuity of the otherwise low-signal triangle of labrum as it becomes confluent with the low-signal articular cartilage (▶ Fig. 10.4).

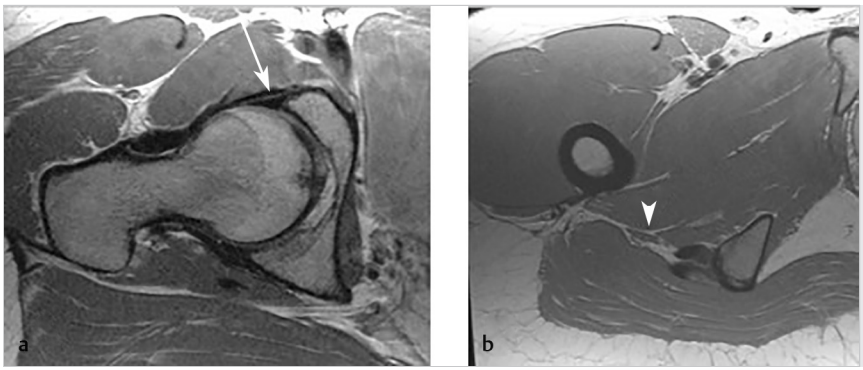


Fig. 10.3 An axial fast spin-echo (FSE) image of the hip demonstrating **(a)** the acetabular labrum (*arrow*) and **(b)** more distally, the sciatic nerve (*arrow head*).

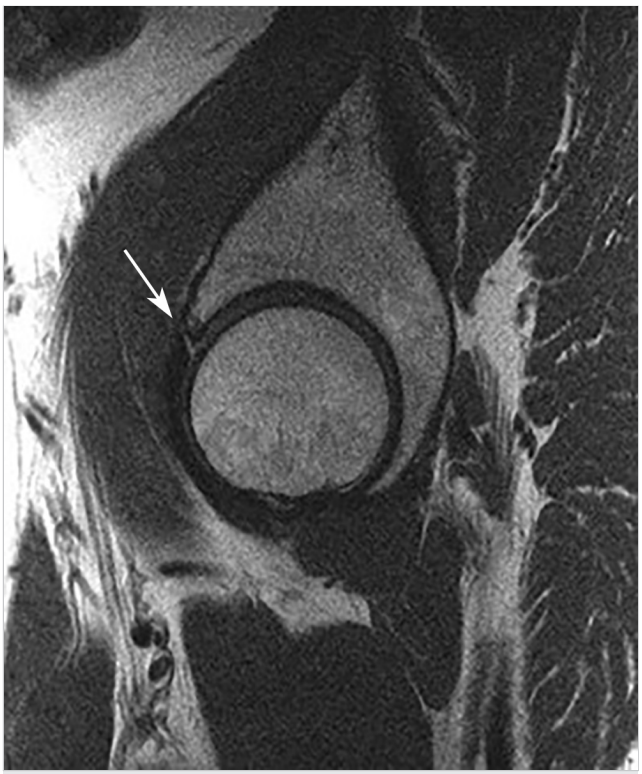


Fig. 10.4 A sagittal fast spin-echo (FSE) image of the hip showing a nondisplaced anterior labral tear (*arrow*). (Source: © 2009 Hospital for Special Surgery, New York, NY.)

Next we examine the ligamentum teres, the joint capsule, the shape of the femoral neck, and the presence/absence and quality of joint effusion. We then evaluate the extra-articular anatomy, including the spine, SI joints, hamstrings, muscles, pubis, and presence/absence of hernia. This survey also takes into account any extra-articular impingement and the global shape of the hip (i.e., acetabular under- or overcoverage).

10.4 Traumatic Conditions and the Athletic Hip

10.4.1 Acetabular Labral Tears

The fibrocartilaginous labrum (type I collagen) appears as a triangular, low signal structure that surrounds the rim of the acetabulum.[8] It is contiguous with the transverse acetabular ligament inferiorly. The labrum is thought to play an important role in maintaining hydrostatic fluid pressure in the joint,[9] and labral pathology is not only a cause of substantial hip pain (▶ Table 10.1) but is also associated with the development of osteoarthritis.[10] The labrum is thickest superiorly and posteriorly and widest anteriorly and superiorly. The joint capsule attaches to the acetabular rim adjacent to the labrum; however, at the superior aspect of the joint, the capsule inserts a few millimeters above the labrum, which creates a sublabral recess or sulcus.[11] Free nerve endings and nerve end organs have been identified in the labrum, most abundantly in the anterosuperior and posterosuperior zones, and they may be involved in pain and proprioception.[12,13] The blood supply to the acetabular labrum is poor overall but there are regional variations, with significantly more vascularity on the capsular side than the articular side.[14] Detachment of the posterior labrum should raise suspicion for a posterior subluxation event.

Most (75%) labral tears occur anterosuperiorly and on the articular side.[15] Labral tears are identified by the presence of increased signal intensity (often equivalent to that of synovial fluid) on fat-suppressed images through the substance of the labrum (▶ Fig. 10.4). Partial or complete detachment is characterized by fluid-matching signal undermining the attachment between the labrum and acetabulum. The presence of paralabral cysts should prompt careful inspection of the labrum to discern the often subtle splits that serve as the origin for these cysts (▶ Fig. 10.5). There is a high rate of asymptomatic labral tears (33–69%) found on MRI in young adults,[16,17] so diagnosis of a labral tear must be paired carefully with history and physical examination. In patients older than 50 years with hip pain, 73% have labral tears and 93% have some labral pathology,[18] whereas in pediatric patients (aged 2–18 y) the rate of asymptomatic labral tear is very low (1.4%).[19]

To optimize outcomes, the underlying cause of the labral injury must be identified. In addition to degenerative labral tears associated with arthritis, there are at least five causes of labral tears[20]:

- Femoroacetabular impingement (FAI).
- Traumatic labral injury (dislocation/subluxation).
- Atraumatic capsular laxity (hypermobility).
- Psoas impingement.
- Dysplasia.

Several classification systems have been used for labral tears, two of which are described here. In a cadaveric study, Seldes et al[15] determined histologically that there were two distinct types of tears (▶ Fig. 10.6):

- Detachment of the labrum from the hyaline cartilage at the transitional zone (type 1).
- Cleavage of variable depth within the substance of the labrum (type 2).

In another study, Lage et al[21] developed an arthroscopic classification based on morphology:

- Radial flap tear.
- Radial fibrillated tear.
- Longitudinal peripheral tear.
- Unstable tear.

Blankenbaker et al[22] found little correlation between MR arthrographic appearance of labral tears and the Lage arthroscopic classification and suggested that morphologic descriptions are the best way to describe tears on imaging. Ilizaliturri et al[23] described a zone method for classifying intra-articular pathology used during hip arthroscopy (▶ Fig. 10.7), which they found to be more reproducible than the clock-face method of describing tears at the time of surgery.

Two potential pitfalls may occur when interpreting noncontrast MRI and MR arthrography studies, potentially leading to an inaccurate diagnosis of a labral tear. First, there is a normal variant of the posteroinferior sublabral groove, which is a relatively common occurrence and typically seen on axial images.[11]

Table 10.1 Differential diagnosis of hip pain

Primary labral pathology	**Nonmusculoskeletal causes**
Femoroacetabular impingement	Genitourinary
Laxity	Spine
Trauma	Psoas muscle abscess
Dysplasia	Hernia
Degenerative	Endometriosis
	Ovarian cyst
	Peripheral vascular disease
Primary chondral	**Bone etiology**
Lateral impact	Subchondral fractures
Subluxation/dislocation	Other fractures
Osteonecrosis	
Loose bodies	
Degeneration	
Primary capsule	**Synovial proliferative disorders**
Laxity	Pigmented villonodular synovitis
Adhesive capsulitis	Synovial chondromatosis
Synovitis/inflammation	Chondrocalcinosis
Extra-articular	**Infectious/tumor/metabolic**
Snapping hip (internal/external)	Septic arthritis
Trochanteric bursitis	Osteomyelitis
Ischial bursitis	
Psoas bursitis/subiliacus bursitis	**Neoplasms**
Osteitis pubis	Benign bone and soft tissue
Core muscle injury (formerly sports hernia or athletic pubalgia)	Malignant bone and soft tissue
Deep gluteal syndrome	Paget disease
Sacroiliac joint	Primary hyperparathyroidism
Tendinitis	Metastatic bone disease
Hip flexor	
Adductor	
Abductor	
Gluteus medius tear Trochanteric impingement Hamstring strain/tear Calcific tendonitis	
Inflammatory	**Systemic**
Rheumatoid arthritis	Polyarticular
Reiter syndrome	Reflex sympathetic dystrophy
Psoriatic arthritis	Complex regional pain syndrome
Bursitis	Hormonal

Source: Reproduced with permission from Shindle MK, Ranawat AS, Kelly BT. Diagnosis and management of traumatic and atraumatic hip instability in the athletic patient. Clin Sports Med 2006;25:309–326. Reprinted by permission.

The clinician should recognize that this location is distinct from that of most labral tears. The second variant is a superior sublabral recess at the anterosuperior margin of the acetabulum, which has been described in the presence of mild acetabular dysplasia.[24]

The high prevalence of labral tears in the anterosuperior labrum is speculated to be secondary to decreased vascularity in this zone,[15,20] as well as the collagen content and fiber orientation in this area.[25] Kalhor et al[26] found that the blood supply to the labrum originates from the periacetabular vascular ring, which then travels toward the free edge of the labrum, traversing the osseolabral junction on the capsular side. These findings supported those of previous studies that identified the main blood supply to the acetabular labrum as located on the capsular surface.[14,15] In examination of fetal labra, Turker et al[25] found a parallel orientation of collagen fibers to the chondrolabral junction in the anterosuperior quadrants and a perpendicular orientation of denser collagen in the posterior inferior quadrant. Such studies have important implications for labral surgery in terms of determining the type and location of tears that would be amenable to successful repair, as well as avoiding detaching the labrum when performing a debridement or repair.



Fig. 10.5 A sagittal fast spin-echo (FSE) image of the hip showing paralabral cysts (*arrow*) associated with an anterior labral tear and full-thickness cartilage loss of the anterior acetabulum. (Source: © 2009 Hospital for Special Surgery, New York, NY)

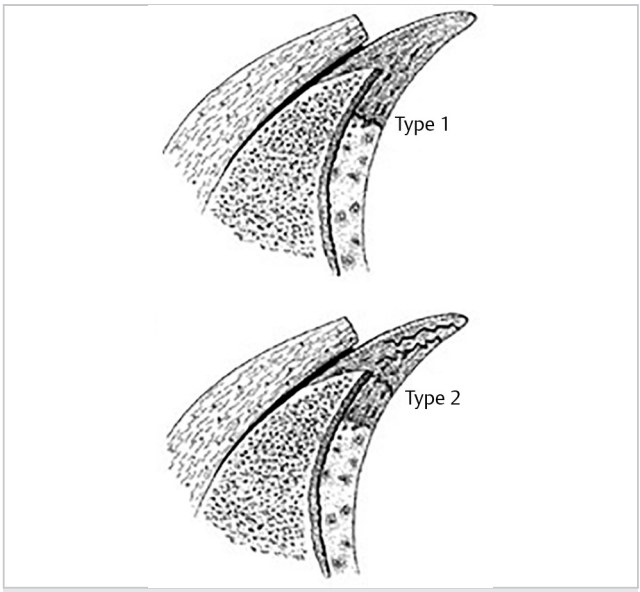


Fig. 10.6 Seldes type-1 and type-2 labral tears: detachment of the labrum from the hyaline cartilage at the transitional zone and cleavage of variable depth within the substance of the labrum, respectively. (Reproduced with permission from Seldes RM, Tan V, Hunt J, Katz M, Winiarsky R, Fitzgerald RH, Jr. Anatomy, histologic features, and vascularity of the adult acetabular labrum. Clin Orthop Relat Res. 2001(382):232–240.[15])

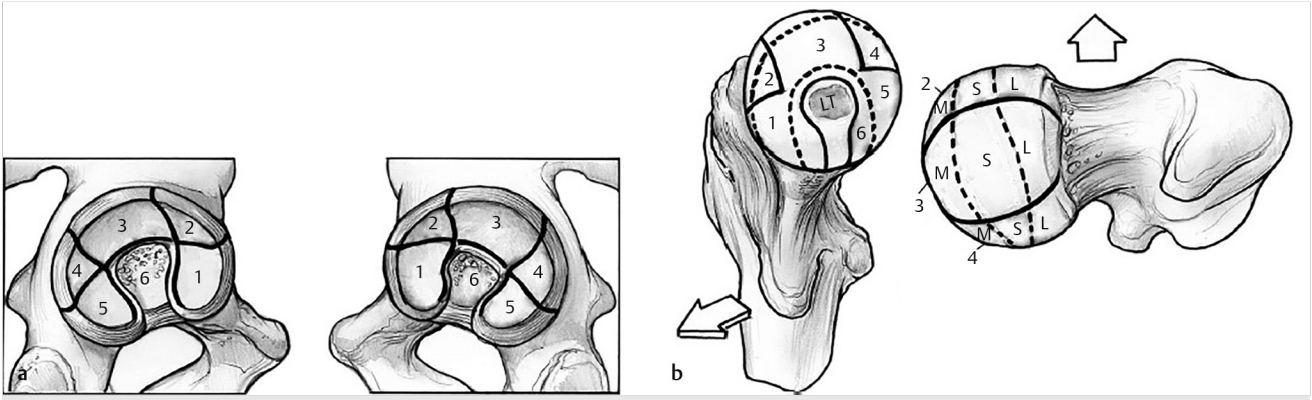


Fig. 10.7 Zone classification. **(a)** An artist's depiction of the left and right hemipelvis. The acetabulum has been subdivided into six zones: 1, anterior-inferior; 2, anterior-superior; 3, central-superior; 4, posterior-superior; 5, posterior-inferior; and 6, acetabular notch. **(b)** An artist's depiction of the left and right proximal femurs. (*Left*) The frontal view of the femoral head is divided into six zones around the acetabular fossa (there is a *dotted line* around the ligamentum teres [*LT*]): 1, anterior-inferior; 2, anterior-superior; 3, central-superior; 4, posterior-superior; 5, posterior-inferior; and 6, area around the ligamentum teres. (*Right*) The superior view shows the three subdivisions of zones 2, 3, and 4: medial (*M*), superior (*S*), and lateral (*L*). (Reproduced with permission from Ilizaliturri VM Jr, Byrd JWT, Sampson TG, et al. A geographic zone method to describe intra-articular pathology in hip arthroscopy: cadaveric study and preliminary report. Arthroscopy 2008;24:534–539.[23])

10.4.2 Femoroacetabular Impingement

Femoroacetabular impingement (FAI) is a well-described pathologic condition that is now understood to lead to hip degeneration and osteoarthritis through progressive chondral and labral injury.[27,28,29,30,31,32] FAI is caused by two main types of mechanism: "cam" and "pincer." Most FAI cases represent a combination of both mechanisms (▶ Fig. 10.8).[27] Although the cause of FAI is not completely understood, subclinical, slipped, capital femoral epiphysis has been theorized to lead to the femoral neck deformity,[33] and more recently it has been suggested that increased loading of the hip during late childhood and early adolescence plays a role in the development of the head-neck junction deformity.[34,35,36] A genetic component has also been suggested; Pollard et al[37] found that the siblings of a patient with FAI had a relative risk of 2.0 of having a rim deformity ($p = 0.001$) and a relative risk of 2.8 of having a cam deformity ($p < 0.00001$).

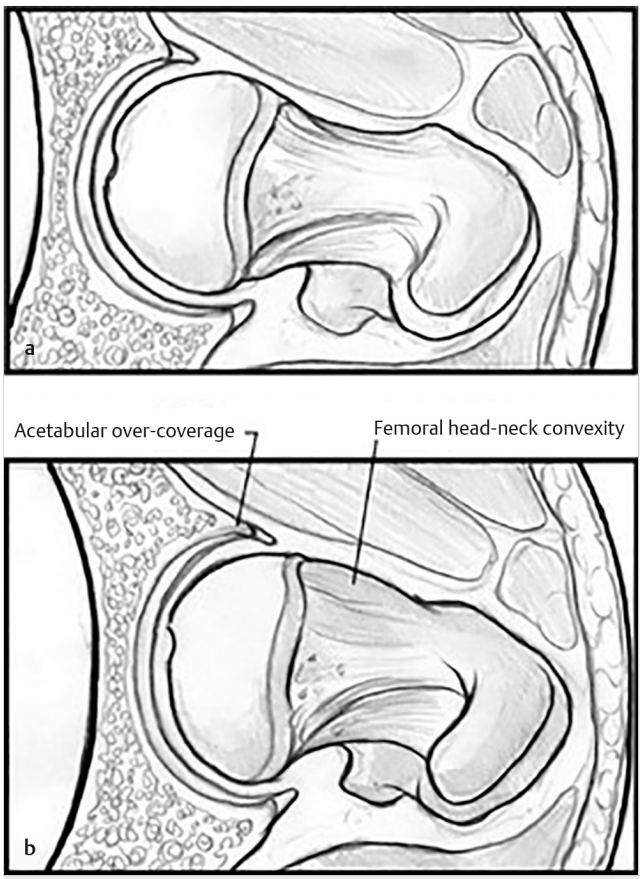


Fig. 10.8 Diagrammatic representation of **(a)** a normal acetabulum and femur morphology and **(b)** combined femoroacetabular impingement with reduced head-neck offset (cam mechanism) and excessive anterior overcoverage (pincer mechanism). (Reproduced with permission from Michael K. Shindle, Bryan T. Kelly, Luis E. Moya, and Douglas N. Mintz. The hip. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:147–163.)

Repetitive microtrauma from impingement during hip flexion and internal rotation leads to degeneration of the acetabular labrum and articular cartilage and is a source of hip pain.[29,31,32] MRI is an excellent modality to evaluate FAI because it permits early detection of the disease process and can identify labral tears, cartilage damage, subchondral cysts, and underlying subtle anatomic variations of the femoral head-neck junction or acetabulum associated with cam or pincer impingement. There is, however, a relatively high incidence of FAI in asymptomatic individuals, so careful correlation of MRI and clinical examination findings is warranted. A recent systematic review found the prevalence of asymptomatic cam deformity was 37% (55% in athletes, 23% in the general population); and the prevalence of asymptomatic pincer lesion was 67%.[38]

Cam-type FAI involves shear forces created by the nonspherical portion of the femoral head-neck junction impinging on the acetabulum, resulting in a characteristic pattern of cartilage loss over the anterosuperior weight-bearing portion of the acetabulum and, eventually, possible tear or detachment of the principally uninvolved labrum (▶ Fig. 10.9).[31,39] MRI can detect subtle anatomic variations of the femoral head-neck junction, labral pathology, adjacent cartilage damage, and subchondral cysts. Typical findings include any of the following[31,39]:

- High signal of the anterior joint margin extending to the tidemark.
- Intermediate-signal labral degeneration.
- Labral tears with a linear or diffuse increased signal intensity.
- Defects, fissures, or delamination in the articular cartilage.
- Femoral head/neck cysts.

Typical findings on fast spin-echo (FSE) may also include abnormal contour of the acetabular labrum and a cam lesion. Decreased offset of the femoral head-neck junction can be detected on conventional radiographs or MRI as a prominent lateral extension of the femoral head at the step-off to the adjacent femoral neck.[31] Other features that can be present on MRI are the following:

- Congruent but nonspherical head.
- Small head-to-neck ratio.
- Short neck.

As part of the multiplanar assessment of FAI, oblique axial MRI sequences can be helpful in defining the abnormal osseous neck-shaft offset. In those images, the segment prescription is parallel to the neck-shaft angle of a coronal image, passing directly through the center of the femoral head. The α angle, measured according to the method of Nötzli et al,[40] can help define the anterior margin of the waist of the femoral neck (▶ Fig. 10.10).

Pincer-type FAI involves contact between a normal femoral neck and an abnormal anterior acetabular rim secondary to focal or diffuse acetabular retroversion or protrusion of the femoral head into the pelvis (protrusio acetabuli or acetabular profunda). In this setting, the acetabular labrum fails first, which leads to degeneration and eventual ossification, worsening the overcoverage (retroversion). In patients with a retroverted acetabulum, axial MRI shows increased coverage of the anterior femoral head.[41] This is also observed on radiographs and computed tomography (CT) images. During flexion and internal rotation, the prominent anterolateral edge of the acetabulum leads to impingement and subsequent labral degeneration. Synovial fluid may penetrate into the subchondral bone and lead to subchondral cyst formation.[4142,43] In addition, persistent anterior abutment can result in contrecoup cartilage lesions of the posteroinferior acetabulum. A pincer-type lesion could also result from focal superior acetabular retroversion. Overall, the pincer type has fewer chondral manifestations compared with the deep chondral lesions associated with cam impingement.[44,45] Intralabral ossification can be noted at the anterior and lateral margins of the hip joint.[39]

High rates of FAI are found in patients with hip instability, suggesting that FAI may predispose patients to hip instability because of the morphologic changes of the bone levering the femoral head posteriorly.[46,47] Although MRI has limitations in evaluating for hip instability, > 5-mm widening of the anterior hip joint recess and thinning to < 3 mm of the joint capsule lateral to the zona orbicularis on MR arthrography has been used as an indication of hip instability.[48]

10.4.3 Stress Fractures

Stress fractures, which result from cyclic mechanical stress, can be categorized into one of two groups: fatigue or insufficiency.

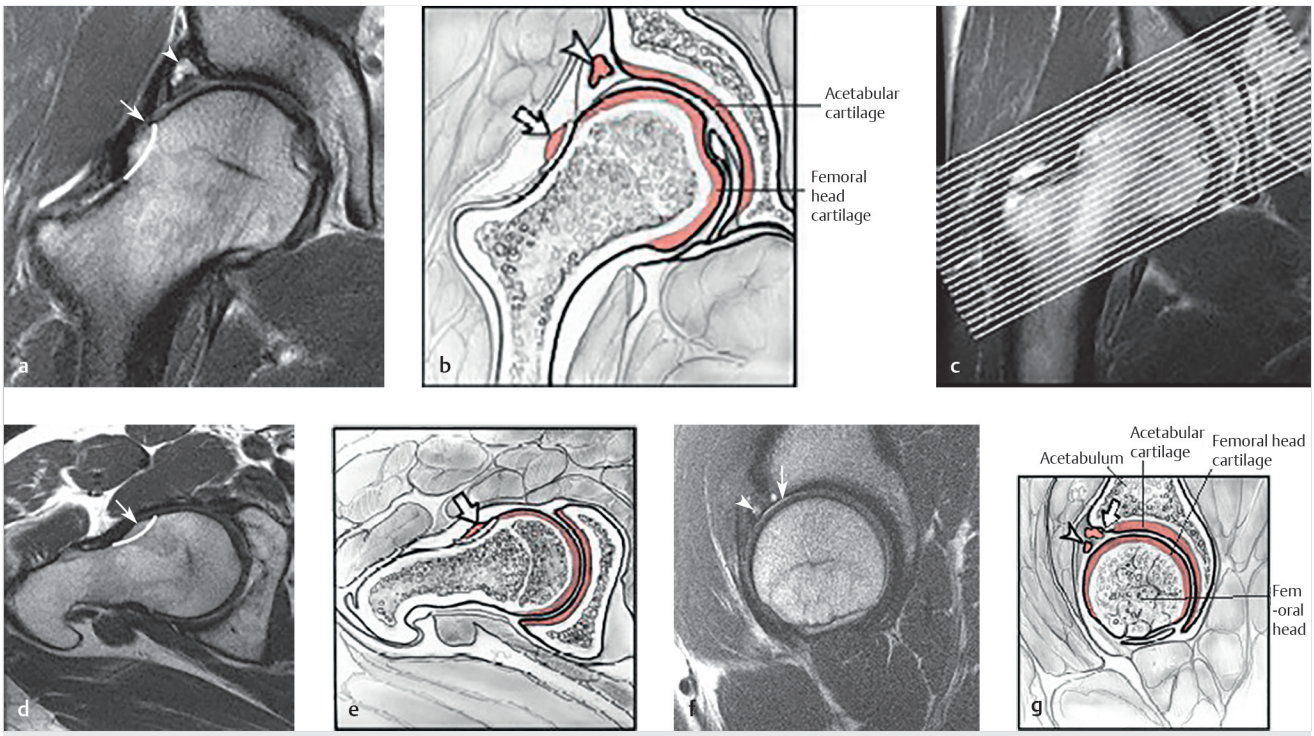


Fig. 10.9 Fast spin-echo (FSE) images and sketches of a 41-year-old patient with combined femoroacetabular impingement of the right hip. **(a)** The coronal with the labral tear and cam lesion **(b)** and associated artist's sketch. **(c)** The series prescription for the oblique axial sequence. **(d)** The oblique axial view **(e)** and associated artist's sketch **(f)** The sagittal MRI sequence **(g)** and associated artist's sketch. (Reproduced with permission from Shindle MK, Foo LF, Kelly BT, et al. Magnetic resonance imaging of cartilage in the athlete: current techniques and spectrum of disease. J Bone Joint Surg Am 2006;88:27–46.)

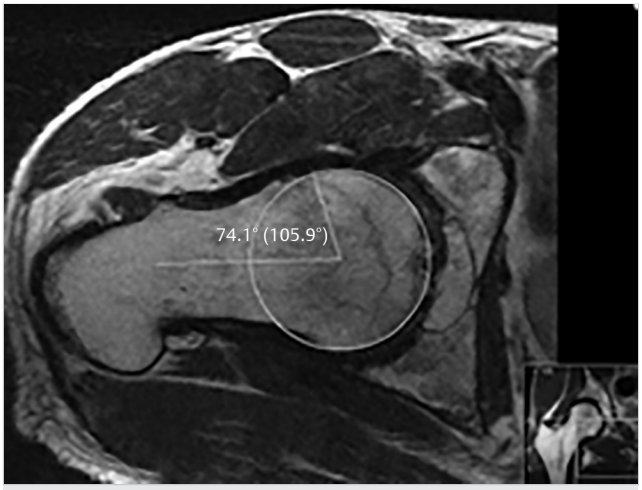


Fig. 10.10 An oblique axial fast spin-echo (FSE) image of the left hip showing an increased α angle in a nonspherical femoral head with a cam lesion. (Source: © 2009 Hospital for Special Surgery, New York, NY.)

Fatigue fractures are caused by abnormal stresses on normal bone; insufficiency fractures result from normal forces in abnormal bone. Military recruits and athletes are more susceptible than the general population to fatigue fractures, given that they occur in normal bone that is exposed to repetitive mechanical overload that overwhelms the physiologic remodeling process.[49] Insufficiency fractures occur more commonly in the elderly secondary to osteoporosis, osteomalacia, or other processes that weaken the bone, permitting osseous failure to occur with fewer loading cycles than in normal bone. Insufficiency fractures in the athletic population have also been described in female athletes, in association with the female athletic triad.[50] Stress fractures can occur on the compression (medial) or the tension (lateral) side of the femoral neck.[51] Compression stress fractures are more common and begin at the inferior cortex of the femoral neck. Tension stress fractures begin in the superior cortex, have a greater propensity to displace, and thus require more aggressive treatment.[52,53] A fracture line with a benign periosteal reaction may be visualized using conventional radiography or CT. Bone scintigraphy is more sensitive but not specific for the diagnosis of femoral neck stress fracture. MRI is both sensitive and specific to the presence of underlying bone marrow lesions, and a linear fracture line associated with the preservation of fatty marrow confirms a diagnosis of a stress fracture.[51,54] Fat-suppressed coronal images show a low signal line amid an area of increased signal intensity, indicating an acute or subacute fracture (▶ Fig. 10.11). MRI is also useful for excluding underlying lesions such as a tumor and marrow disorders that may be the cause of a pathologic stress fracture.

10.4.4 Traumatic Posterior Subluxation/ Dislocation/Lateral Impaction

Traumatic posterior hip dislocation most commonly occurs as a result of a motor vehicle accident when the hip and knee are flexed and the knee impacts the dashboard, producing a posteriorly directed force on the hip. Blanchard et al[55] found that in children who sustained traumatic hip dislocation, a consistent pattern of near complete avulsion of the posterior labrum occurred. In the acute setting, MRI may aid in the diagnosis of the following[56,57,58]:

- Femoral head contusions.
- Chondral injury.
- Labral disruption or entrapment.
- Capsular tear.
- Sciatic nerve injury.
- Intra-articular fragments.
- Ligamentum teres disruption.
- Nondisplaced fractures of the femoral head or acetabulum.

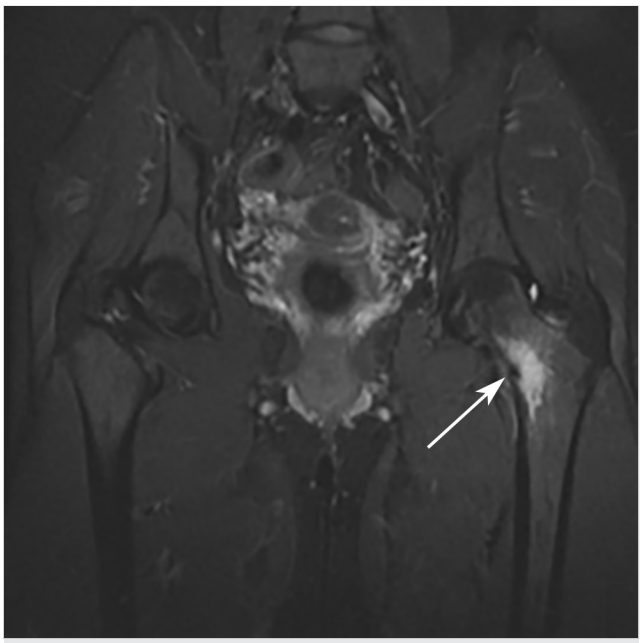


Fig. 10.11 A coronal fat-suppressed image obtained with a body coil technique, showing increased signal intensity (*arrow*) surrounding a stress fracture at the medial aspect of the left femoral neck. (Source: © 2009 Hospital for Special Surgery, New York, NY.)

Hip subluxation can occur with low-energy mechanisms such as a fall onto a flexed hip or knee, and because its presentation is often subtler than hip dislocation, it can be misdiagnosed as a hip strain or sprain. Hip subluxation can, however, lead to substantial morbidity if missed. Traumatic posterior subluxation can occur in contact sports such as football and may result in symptomatic cartilage/labral injury despite a lack of radiographic evidence of dislocation.[59] Hip translation associated with a subluxation event can result in anterior or posterior labrum or posterior wall displacement. The iliofemoral ligament and parafoveal cartilage can also be disrupted, typically during relocation of the hip joint (▶ Fig. 10.12). MRI is useful for monitoring sequelae of the subluxation, including osteonecrosis and femoral head edema (see section "Osteonecrosis").[59,60] Serial MRI can aid in determining when an athlete can return to sport. A direct blow to the greater trochanter in young, otherwise physically fit patients can result in an isolated chondral injury, as described by Byrd et al.[61] MRI can be used to determine the extent of chondral damage and the presence of osteochondral loose bodies.

10.4.5 Muscle Strains and Tears

The most frequent injuries about the groin and hip are muscle strains and tears. These injuries typically affect the myotendinous junction, which is the weakest point of the musculotendinous unit, but tendon avulsion from the osseous attachment can also occur.[62] The most commonly strained muscles around the hip joint are the following[62]:

- Hamstrings (biceps femoris, semimembranosus, and semitendinosus).
- Rectus femoris.
- Adductors.

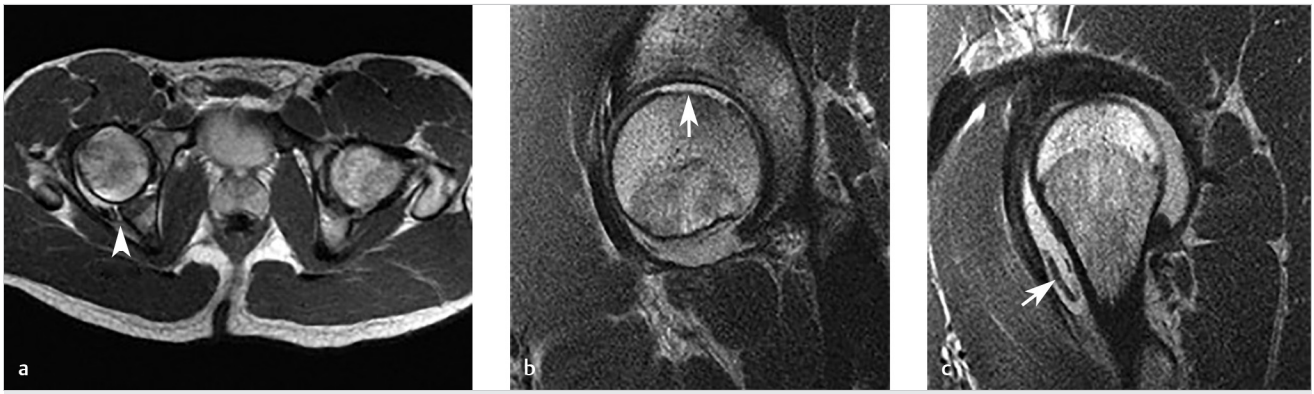


Fig. 10.12 Posterior hip subluxation. Axial body coil **(a)** and sagittal surface coil **(b,c)** fast spin-echo (FSE) images of the hip in an 18-year-old patient with sequelae of posterior hip subluxation. An intact right posterior hip capsule is seen, attached to a posterior wall fracture **(a**, *arrowhead*) and associated with a large, full-thickness chondral shear injury of the femoral head **(b**, *arrow*). Cartilaginous debris **(c**, *arrow*) is seen within the anteroinferior dependent recess of the joint. (Reproduced with permission from Shindle MK, Foo LF, Kelly BT, et al. Magnetic resonance imaging of cartilage in the athlete: current techniques and spectrum of disease. J Bone Joint Surg Am 2006;88:27–46.)

The adductor group is frequently involved, especially in hockey, soccer, and football players. Medial thigh or groin pain is the presenting symptom.[62] The MRI classification is based on the extent of disruption present: on fat-suppressed images, muscle avulsions and tears typically have high signal intensity in areas of hemorrhage or edema. First-degree strains have a minor degree of fiber disruption and a "feathery" pattern of increased signal intensity intermediate to T2-weighted images secondary to interstitial edema and hemorrhage that extends into the adjacent muscle fascicles. Second-degree strains have a partial tear without retraction and often also have a hematoma at the myotendinous junction. Third-degree strains have a complete rupture of the myotendinous unit.[63] Acute tendinous avulsion injuries of the rectus femoris origin at the anterior inferior iliac spine can be seen. This injury occurs most commonly after forceful extension of the hip (▶ Fig. 10.13).[64] Hamstring tears are typically partial-thickness tears, and the biceps femoris is the most frequently injured structure, followed by the semitendinosus and semimembranosus muscles. Injuries typically occur at the musculotendinous junction with the characteristic featherlike edema pattern described above.

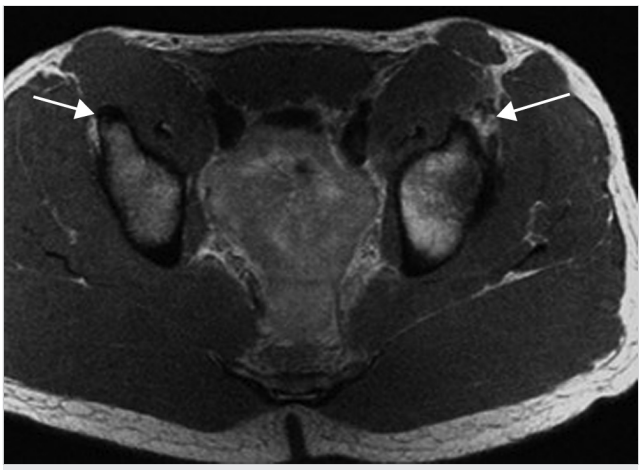


Fig. 10.13 An axial fast spin-echo (FSE) image obtained with a body coil technique, showing avulsion of the proximal portion of the rectus femoris from the patient's left anterior inferior iliac spine (*left arrow*). A normal low-signal tendinous insertion is seen on the right (*right arrow*). (Source: © 2009 Hospital for Special Surgery, New York, NY.)

10.4.6 Apophyseal Avulsion

Apophyseal avulsion injuries characteristically occur in adolescents and are equivalent to muscle strains in adults. During running and jumping activities, the hamstring origins at the ischial tuberosity may be avulsed; these injuries have been referred to as hurdler's fractures.[65] Apophyseal injuries of the pelvis may also involve the rectus femoris or sartorius muscles at the anteroinferior or anterosuperior iliac spines, respectively. In competitive sports that require kicking motions with extreme hip flexion, avulsions of the iliopsoas tendon off the lesser trochanter may be seen.

10.4.7 Athletic Pubalgia/Core Muscle Injury

The term "athletic pubalgia" describes a spectrum of activity-induced pain that occurs at the pubic symphysis. This can include osteitis pubis, adductor, or core muscle injuries.[66] MRI plays a central role in identifying the cause of symptoms in athletic pubalgia. Osteitis pubis is a noninfectious inflammation of the pubic symphysis, which is thought to be related to instability at the pubic symphysis resulting from ligamentous, muscular, or tendinous injury.[67] MRI will demonstrate bone marrow edema at the pubis, fluid within the pubic symphysis joint, or symphyseal widening[68] (▶ Fig. 10.14). Core muscle injuries, or rectus abdominis/adductor aponeurosis injuries, involve the insertion of the rectus abdominis, the origin of the adductor longus, and the periosteum of the pubic tubercle.[69] This injury has been referred to as sports hernia, Gilmore groin, pubic inguinal pain syndrome, or sportsmen's groin.[70] MRI will demonstrate marrow edema at the pubis, a visible defect in the rectus abdominis anterior to the pubic tubercle, detachment of or fluid interspersed within the adductor longus tendon, or a cleft beneath the pubic bone indicating a lesion within the aponeurosis.[69] Injuries can also be present further posteriorly in the pubis and can extend to the adductor brevis.

10.4.8 Ligamentum Teres Tears

The ligamentum teres tightens in hip abduction, medial rotation, and lateral rotation. It is thought to prevent inferior, posterior, and anterior subluxation of the femoral head in abduction, medial rotation, and lateral rotation, respectively.[71] However, its contribution to overall hip function is debated. Tears of the

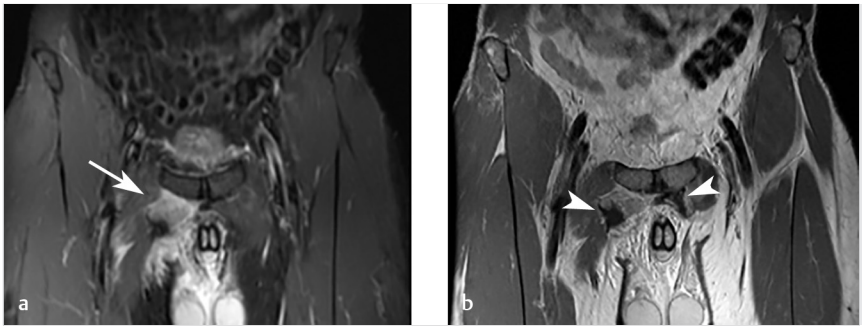


Fig. 10.14 Core muscle injury. **(a)** Right-sided edema at origin of adductor tendons on inversion recovery sequence (*arrow*). **(b)** Coronal proton-density sequence demonstrating the avulsed tendon on the patient's right (*left arrowhead*) and the intact tendon on the left (*right arrowhead*).

ligamentum teres can be a source of hip pain, and some degree of tearing has been identified in 50 to 88% of patients undergoing hip surgery.[72,73] Chahla et al[73] found that the ligamentum teres was frayed or partially torn in 88% of patients undergoing hip arthroscopy and that it was fullytorn in only 1.5%. Complete tears of the ligamentum teres were associated with increased hip laxity and femoral head chondral defects. A higher incidence of atraumatic ligamentum teres tears has been found in athletes who repetitively load the ligament with excessive hip range of motion, such as professional ballet dancers.[74] Ligamentum teres hypertrophy has been noted in asymptomatic elite gymnasts.[75]

10.4.9 Greater Trochanteric Pain Syndrome

Greater trochanteric pain syndrome is a term that encompasses greater trochanteric bursitis, effusion, tears/tendinosis of the gluteus medius and minimus, and external coxa saltans (snapping hip). The major clinical findings are nonspecific, typically with tenderness localized to the lateral aspect of the hip or thigh. Clinically, distinguishing isolated bursal pathologic changes from tendon abnormalities or other soft tissue lesions is difficult; therefore, the differential diagnosis is broad and requires consideration of articular, periarticular, and distant processes as sources of lateral hip pain.[76]

Trochanteric Bursitis

The surface of the greater trochanter has four distinct facets:
- Anterior.
- Posterior.
- Lateral.
- Superoposterior.

The overlying trochanteric bursa is composed of three discrete parts[76]:
- Subgluteus maximus bursa.
- Subgluteus medius bursa.
- Subgluteus minimus bursa.

Excessive friction from the overlying iliotibial band (ITB) is the most likely cause but trochanteric bursitis is often a manifestation of underlying abductor tendon pathology. Thus, the presence of trochanteric bursitis should prompt careful attention to the underlying abductor tendons. Fat-suppressed MR sequences, particularly coronal and axial images, should be evaluated for bursal distention with high signal intensity (▶ Fig. 10.15a).

10.4.10 Abductor Tears

The gluteus medius inserts onto the superoposterior and lateral facets, and the gluteus minimus inserts onto the anterior facet of the greater trochanter. MRI evaluation of the hip in patients older than 40 years often reveals several pathologic processes, including some degree of the following[18,77,78]:
- Cartilage degeneration.
- Degenerative labral lesions.
- Extracapsular abnormalities (including abductor or iliopsoas tendinosis).

Abductor tears have been termed the "rotator cuff tears of the hip" and are highly prevalent in patients with osteoarthritis.[79,80] Coronal and axial MRI sequences should be evaluated for the presence or absence of such tears. Partial-thickness abductor tears show hyperintensity with abnormal morphology or alterations in tendon caliber and signal of the torn tendon fibers, and full-thickness tears show a fluid-filled gap with a retracted tendon (▶ Fig. 10.15b). Some patients have "intractable" symptoms of trochanteric bursitis despite multiple injections and nonoperative treatment. For such patients, MRI may be diagnostic for a gluteus medius or minimus rupture.[81]

10.4.11 Snapping Hip Syndrome (Coxa Saltans)

Snapping hip syndrome, or coxa saltans, is characterized by an audible or palpable snapping of the hip during flexion or extension, occasionally causing pain. The causes of coxa saltans have been divided into three categories[82]:
- Intra-articular.
- Internal (snapping of the iliopsoas tendon against an osseous ridge).
- External (snapping of the ITB or gluteus maximus muscle over the greater trochanter).

It is important to accurately determine the source of the snapping so the appropriate treatment can be given. MRI is useful

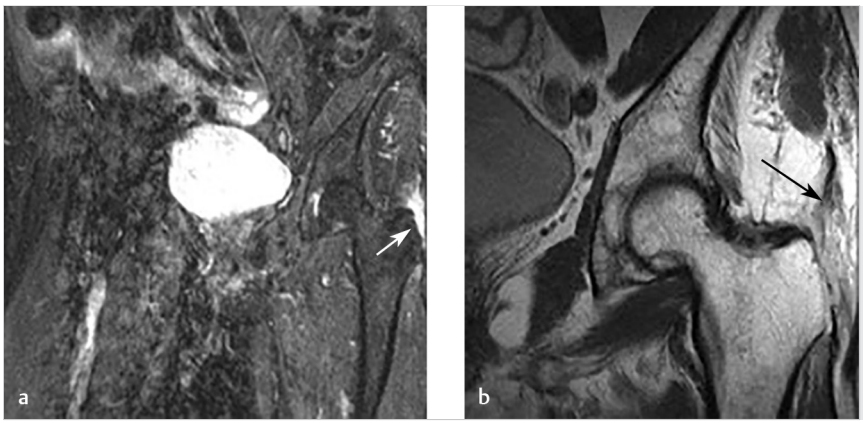


Fig. 10.15 Hip abductor tear. **(a)** A coronal fat-suppressed image of the hip showing a fluid-filled space that should be occupied by the left hip abductors, indicating a tear (*arrow*). **(b)** A fast spin-echo (FSE) image showing the tear with retraction of the gluteus medius tendon (*arrow*) with dehiscence of its insertion from the greater trochanter. (Source: © 2009 Hospital for Special Surgery, New York, NY.)

for excluding other intra- or extra-articular hip abnormalities but its role in diagnosing coxa saltans is limited because of its lack of dynamic capabilities. With internal coxa saltans, fat-suppressed images may display iliopsoas bursa fluid. With external coxa saltans, fat-suppressed images may show a high-signal ITB, trochanteric bursa, and anterior border of the gluteus maximus. Dynamic sonography of the hip is beneficial as an adjuvant to static MRI to confirm the diagnosis.

10.4.12 Deep Gluteal Syndrome

Deep gluteal syndrome (DGS) describes pain in the buttock area, hip, or posterior thigh and/or nondiscogenic radicular pain caused by entrapment of the sciatic nerve within the subgluteal space.[83] The sciatic nerve exits the pelvis through the greater sciatic notch. Although variations exist, the nerve is most often located immediately anterior to the piriformis muscle. Irritation of the piriformis musculotendinous unit can cause compression of the sciatic nerve,[84] which is a subtype of DGS often referred to as "piriformis syndrome." MRI may show an abnormal hypertrophic piriformis muscle and/or hyperintensity of the sciatic nerve on fat-suppressed images, a variant course of the nerve through the piriformis muscle, or a mass effect, resulting in displacement of the muscle from anterior to posterior with a loss of normal muscle striations.[64] Other sources of extrapelvic sciatic nerve compression include fibrous bands interposed with blood vessels, gluteal muscles, hamstring muscles, the ischial tuberosity, the gemelli-obturator internus complex, or quadratus femoris or ischiofemoral pathology.[83,85] MRI is useful to carefully evaluate the sciatic nerve and the structures that could be causing the patient's DGS.

10.5 Degenerative Conditions

10.5.1 Osteoarthritis

Osteoarthritis is characterized by articular cartilage loss with hip joint space narrowing; the incidence of this process increases with age. The hip is one of the most common sites for osteoarthritis,[86] and there are several conditions that lead to osteoarthritis, including the following[31]:

- FAI.
- Previous slipped capital femoral epiphysis.
- Developmental dysplasia of the hip.
- Legg-Calvé-Perthes disease.
- Trauma.
- Osteonecrosis.
- Paget disease.
- Other anatomic variants.

It is now well accepted that FAI is a precursor for the development of hip osteoarthritis.[27,28,29,30,31,32] The appearance of osteoarthritis on MRI is similar to that on conventional radiographs, including the following:

- Hypointense subchondral bone of the femoral head and acetabulum.
- Joint space narrowing (thinning of cartilage).
- Osteophyte formation.
- Subchondral cysts.
- Deformity of the femoral head or acetabulum.

Osteoarthritis may also be associated with labral tears, paralabral cysts, synovitis, or osteonecrosis of the femoral head.[87] With advancements in quantitative MRI, there is increased focus on evaluating articular cartilage thickness and ultrastructural integrity in the setting of FAI or prior hip injury.[88,89]

10.5.2 Dysplasia

Acetabular dysplasia describes acetabular undercoverage of the femoral head. Patients may experience hip instability as a result of increased forces on the secondary, soft tissue stabilizers of the hip joint, and abnormal forces on articular cartilage can lead to the development of osteoarthritis.[90,91,92,93] Hip dysplasia is classically diagnosed on the basis of an anteroposterior radiograph of the pelvis, in which the lateral center edge angle is < 25 degrees. Most dysplastic hips have a component of acetabular anteversion, but 15 to 20% have acetabular retroversion.[94,95] Acetabular retroversion has been associated with early osteoarthritis, and if there is concomitant femoral retroversion, the affect seems to be additive in leading to early osteoarthritis.[96] MRI can demonstrate cartilaginous and soft tissue changes that may be associated with dysplasia. These can include a hypertrophied labrum and/or ligamentum teres.[97]

10.6 Other Pathologic Conditions

10.6.1 Osteonecrosis

Osteonecrosis (also called avascular necrosis) of the femoral head is the cellular necrosis of the bone and bone marrow elements that comprise part or all of the femoral head. The cause of osteonecrosis is unknown, or "idiopathic," in up to 30% of cases[98] but some of the known predisposing causes including the following[99]:

- Corticosteroid use.
- Collagen vascular disease.
- Trauma (dislocation of the hip, femoral neck fracture).
- Hemoglobinopathies (e.g., sickle cell disease).

MRI has been shown to be more sensitive than conventional radiography, nuclear scintigraphy, or CT for the early detection of osteonecrosis.[99,100,101,102] Besides aiding in early detection, MRI is useful for differentiating osteonecrosis from other pathologic processes and for defining the extent of osteonecrosis, which is a predictor of the likelihood of femoral head collapse.[103,104] Although sometimes difficult, it is especially important to distinguish between osteonecrosis and subchondral insufficiency fractures. Subchondral insufficiency fractures often present as acute onset of severe pain, and MRI will show the fracture parallel and adjacent to the subchondral plate, following the contour of the femoral head.[39]

Osteonecrosis of the femoral head has a characteristic MRI appearance: a low-signal, crescent-shaped rim in a subchondral location that corresponds to the interface between normal and ischemic bone (▶ Fig. 10.16). A second rim of increased signal that is pathognomonic of osteonecrosis typically is present on fat-suppressed images.[101,103]

The presence of a band of low signal intensity on T1-weighted images or of the "double-line" sign on T2-weighted or fat-suppressed proton-density images is each reported to be an



Fig. 10.16 A coronal fast spin-echo (FSE) image of the left hip showing a region of osteonecrosis at the weight-bearing surface of the femoral head with a classic serpentine line of demarcation from the adjacent normal bone. (Source: © 2009 Hospital for Special Surgery, New York, NY.)

early specific sign of nontraumatic osteonecrosis of the femoral head.[101,102] The double-line sign, which appears as a low-signal-intensity peripheral line and a parallel inner line of high signal intensity, can be observed in up to 80% of lesions.[31] These MRI abnormalities represent the reactive interface between necrotic and viable bone.[101,105] MRI may also show a joint effusion on fat-suppressed sequences.

In the presence of a posterior hip subluxation, MRI is useful for detecting osteonecrosis. In the acute setting, MRI is not an accurate predictor of osteonecrosis, and a repeat study should be obtained at 6 weeks. If there is no evidence of osteonecrosis, patients may safely return to sports activity. However, if there is evidence of osteonecrosis, patients are at increased risk for subsequent collapse and joint degeneration.[59]

Limited MRI examination of the hips (without extra time required for scout images) can be obtained to detect femoral head osteonecrosis and to quantify of the size of the lesion.[106] Several methods have been proposed to measure the extent of osteonecrosis for prognosis determination and treatment planning. Lafforgue et al[107] measured the α angle on the coronal T1-weighted images, selecting the largest area of necrosis and determining the angle formed at the center of the femoral head. An angle of > 75 degrees corresponded with a poor prognosis, and an angle of < 46 degrees corresponded to a satisfactory clinical response. Koo and Kim[108] measured the angle formed from the center of the femoral head on midcoronal and midsagittal MR images to quantify the size of the lesion objectively. Cherian et al[109] modified this method for quantifying the extent of osteonecrosis of the femoral head by using the coronal and sagittal MR images with maximal femoral head involvement. They concluded that that method was a reproducible and reliable way of determining the extent of osteonecrosis.

A low percentage of involvement of the femoral head weight-bearing surface has been well correlated with favorable outcomes in patients with early stages of the disease[107] and the unlikelihood of femoral head collapse after core decompressive surgery.[110] In one study, femoral head collapse did not occur when MRI showed that < 25% of the weight-bearing surface was involved.[110] However, when > 50% of the surface was involved, 87% of hips had femoral head collapse.[110]

MRI is also useful for identifying and localizing the necrotic lesion before performing a core decompression procedure. It is not uncommon to see cartilage loss and geode formation on the acetabular side as a result of osteonecrosis of the femoral head.[107] This finding is important because it may change the treatment approach—for example, total hip replacement rather than femoral head resurfacing or core decompression.

10.6.2 Idiopathic Transient Osteoporosis

Idiopathic transient osteoporosis is a rare and often unrecognized cause of hip pain that is usually seen in healthy young adults. The condition most commonly affects women in the third trimester of pregnancy and middle-aged men. Symptoms often present insidiously, with pain in the affected hip, and then progress to gait disturbance and functional disability. Transient osteoporosis of the hip should be considered after more common causes of hip pain (such as infection, stress fractures, or malignancy) are ruled out. Conventional radiographs typically show focal osteopenia of the femoral head and neck with loss of cortical continuity, although this finding may be subtle and is frequently missed. Fat-suppressed MR images show diffuse high signal intensity compatible with edema in the femoral head and proximal femoral neck and a small hip joint effusion, which are the classic findings of transient osteoporosis of the hip. However, this diffuse edema pattern may be related to a subtle subchondral fracture (▶ Fig. 10.17).[111,112] This entity can be differentiated from osteonecrosis of the hip, which shows signal changes at the weight-bearing surface of the femoral head and is not as diffuse as that seen with transient osteoporosis of the hip. Most cases resolve spontaneously within 6 to 12 months. However, subchondral fractures leading to femoral head collapse have been reported.[31,113] Therefore, patients should be treated with protected weight-bearing and monitored carefully with repeat MRI until the edema is resolved and the clinical symptoms subside.

10.7 Inflammatory Disorders

Rheumatoid arthritis (RA) is a systemic autoimmune inflammatory disorder that primarily affects the small joints of the hands and feet but may involve the larger weight-bearing joints, including the hip. Over the past decade, in concert with an increase in antirheumatic drug consumption, there has been a decrease in the number of RA patients undergoing hip replacements, indicating a decline in the effect of RA at the hip.[114] MRI is useful for evaluating erosions, effusion, and synovial pannus formation in patients who are affected at the hip.[115] The

synovial proliferation is typically a mass-like prominence of tissue within the joint that is hyperintense on fat-suppressed sequences.[31] The pannus may be heterogeneous or homogeneous in appearance, and pannus hemorrhages can form with areas of low signal hemosiderin.[115] As with osteoarthritis, the MR findings of inflammatory arthritis correlate with the radiographic appearance, including any of the following[115]:

- Symmetric joint space narrowing.
- Periarticular erosions.
- Juxta-articular osteoporosis.
- Protrusio acetabuli.

Other findings may include the following[31]:

- Hyperintense joint effusion.
- Joint capsule distention.
- Hyperintense marrow edema.

Ankylosing spondylitis is another form of inflammatory arthritis. It most often involves the axial skeleton, and the hip is affected in 24 to 36% of patients, with 9 to 22% demonstrating radiographic evidence of hip arthritis.[116,117,118] Ankylosing spondylitis can have symmetric involvement in the hips but it can be distinguished from RA by its characteristic involvement of the SI joints when the hips are involved.[115] MRI is useful in identifying active inflammatory lesions of the spine and SI joints, which can aid in the diagnosis of this condition.[119]

10.8 Synovial Disorders

Synovial proliferative disorders are not uncommon, so careful scrutiny of the synovium on MRI is warranted. In the hip, negative intra-articular pressure and the tight capsule may result in more prominent erosions than those associated with synovial proliferative disorders in other joints such as the knee, which has several patulous synovial recesses.[31] Synovial chondromatosis is a monoarticular synovium-based cartilage metaplasia that commonly involves the hip joint.[31,120] MRI will show intracapsular fragments of intermediate signal intensity, and this can progress to a "cast" of synovium when the disease becomes confluent (▶ Fig. 10.18).[39] Ferro et al[121] found that patients with synovial chondromatosis do well after arthroscopic synovectomy.

Pigmented villonodular synovitis presents in either a nodular form or a diffuse form. Hemosiderin degradation products creates a low signal intensity that can be helpful in confirming the



Fig. 10.17 Subchondral fractures of the hip. **(a)** A coronal fast spin-echo (FSE) image with fat suppression showing diffuse and extensive high signal of the left femoral head without a demarcated focus (*arrow*). Because there is no evidence of demarcation defining osteonecrosis, the diagnosis is consistent with transient osteoporosis of the hip. **(b)** Most cases of transient osteoporosis actually represent subchondral femoral head stress fractures, which can be appreciated on this coronal FSE image (*arrow*). (Source: © 2009 Hospital for Special Surgery, New York, NY.)

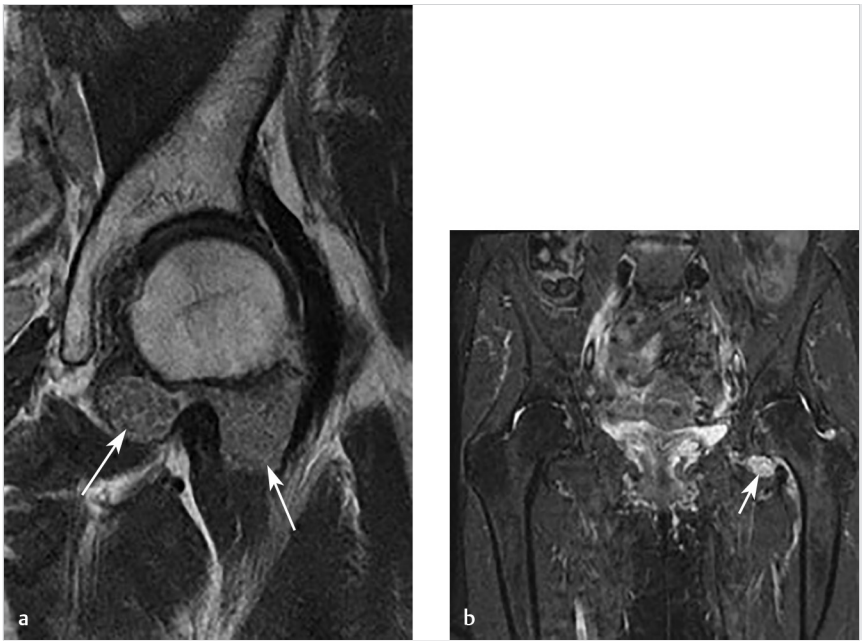


Fig. 10.18 Synovial osteochondromatosis. **(a)** A coronal fast spin-echo (FSE) image of the left hip showing synovial osteochondromatosis with cartilaginous bodies in the inferior capsule (*arrows*). **(b)** A coronal fat-suppressed FSE image of the pelvis also showing detached cartilaginous bodies (*arrow*). (Source: © 2009 Hospital for Special Surgery, New York, NY.)



Fig. 10.19 Pigmented villonodular synovitis. A coronal fast spin-echo (FSE) image of the left hip showing pigmented villonodular synovitis with hemosiderin-laden synovium (*arrow*). (Source: © 2009 Hospital for Special Surgery, New York, NY.)

diagnosis (▶ Fig. 10.19).[122] In its nodular form, it is seen as an intermediate to low signal intensity focus of solitary, intracapsular soft tissue mass within the anteromedial recess or bare area.[31,39] The diffuse form may be characterized by rapid cartilage loss. MRI can help quantify the amount of intra- and extracapsular disease, which is helpful in preoperative planning.[39]

10.9 Postoperative MRI Findings

MRI evaluation of the hip after treatment for FAI with femoral osteochondroplasty, acetabular rim resection, and/or labral repair can aid in diagnosing the cause of persistent or recurrent pain after surgery. Considerations for postoperative MRI include using a gradient-echo sequence in patients with metallic implants[123] and accounting for normal postoperative changes in the bone and soft tissues.

The most common reason for revision surgery is residual structural deformity.[124] MRI can show any residual head-neck junction offset and the extent of femoral osteochondroplasty. Insufficiency fractures of the femoral neck after osteochondroplasty have been reported at an average of 4.4 weeks postoperatively and at a frequency of 1.9% at one institution.[125] These fractures are typically nondisplaced and can be seen on fat-suppressed MRI as a low-intensity signal line surrounded by bone edema.[123] Other postoperative sources of pain that can be identified on MRI include[123]:

- Intra-articular adhesions.
- Progressive joint degeneration.
- Retear of the labrum.
- Capsular incompetence.
- Heterotopic ossification.

Intra-articular adhesions typically occur between the labrum and the joint capsule or between the joint capsule and the site of femoral neck osteochondroplasty.[126] Assessing a retear of the acetabular labrum can be challenging because labral debridement will cause a persistent abnormal intralabral signal secondary to abnormally mobile water in the degenerative fibrocartilage. Blankenbaker et al[127] found that the diagnosis of recurrent tear was best achieved when MR arthrography demonstrated the following: a new line extending to the labral surface, a distorted or enlarged labrum, or a new paralabral cyst. They also found that after labral tear excision, labra appeared shortened on MRI. Capsular incompetence after surgery could be iatrogenic if the capsulotomies used in hip arthroscopy are not closed or it could be the result of capsular dehiscence. MRI will demonstrate an anterior capsular defect with fluid extravasation beneath the iliopsoas bursa.[123] Bedi et al[128] described the incidence of heterotopic ossification after hip arthroscopy as 4.7%, most often in male patients whose surgical procedure included a capsular cut and osteoplasty. Introducing indomethacin into the postoperative prophylaxis protocol reduced the incidence from 8.3% in patients receiving naproxen only to 1.8%.[128]

10.10 Brief High-Yield Review

- The normal hip anatomy can be divided into the osteochondral layer, the capsulolabral layer, the muscular layer, and the neural layer. Sagittal MRI is especially helpful in evaluation of the anterior acetabular labrum. Coronal MRI gives the best view of the weight-bearing cartilage layer of the acetabulum. Axial MRI allows for evaluation of the sciatic nerve.
- Most labral tears occur anterosuperiorly. There is a high incidence of asymptomatic tears, so a tear found on imaging must be clinically correlated.
- MRI is sensitive and specific in identifying stress fractures. Other conditions that may present with increased signal intensity in the femoral head/neck include transient osteoporosis of the hip, which appears as diffuse high signal intensity, or infection.
- Posterior hip subluxation/dislocation and lateral impaction can result in chondral and labral damage, as well as femoral head and sciatic nerve contusions or intra-articular fragments.
- Muscle strains and tears are the most frequent injuries around the groin and hip and can include injury to the hamstrings, adductors, abductors, and core muscles.
- Lateral hip pain can result from greater trochanter pain syndrome, which encompasses trochanteric bursitis, abductor tears, and snapping hip syndrome.
- DGS is a nondiscogenic pain syndrome resulting from entrapment of the sciatic nerve in the subgluteal space.
- FAI is caused by “cam,” “pincer,” or a combination of “cam” and “pincer.” Repetitive microtrauma from impingement of these lesions leads to degeneration of the cartilage of the hip joint.
- MRI is the most sensitive method of imaging to diagnose osteonecrosis, or the necrosis of the bone and bone marrow elements, of the femoral head. The characteristic MRI finding is the “double-line” sign on T2-weighted or fat-suppressed proton-density images.

- Inflammatory conditions of the hip include ankylosing spondylitis and RA. Ankylosing spondylitis often has SI joint involvement. RA on MRI will present with a synovial proliferation. Other synovial proliferative disorders include synovial chondromatosis, which presents on MRI with intracapsular fragments of intermediate signal intensity, and pigmented villonodular synovitis, which is seen on MRI as nodular or diffuse intermediate to low signal, intracapsular, soft tissue masses.
- Postoperative MRI can be obtained to assess for recurrent or persistent pain after surgical treatment of FAI. MRI is useful to assess for residual structural deformity, new labral tears, insufficiency fractures of the femoral neck, capsular incompetence, or heterotopic ossification.

10.11 Common Clinical Questions

1. What is the best MRI sequence when looking for a labral tear?
   a) Coronal.
   b) Sagittal.
   c) Axial.

2. What is the prevalence of asymptomatic FAI?
   a) Asymptomatic CAM, 67%; asymptomatic pincer, 37%; asymptomatic labral tear, 15 to 35%.
   b) Asymptomatic CAM, 37%; asymptomatic pincer, 67%; asymptomatic labral tear, 33 to 69%.
   c) Asymptomatic CAM, 67%; asymptomatic pincer, 37%; asymptomatic labral tear, 55 to 79%.
   d) Asymptomatic CAM, 37%; asymptomatic pincer, 67%; asymptomatic labral tear, 55 to 79%.

3. How should I manage an athlete with suspected hip subluxation?
   a) If no edema on initial MRI, allow to return to sports when asymptomatic.
   b) If evidence of femoral head edema, proceed immediately to core decompression.
   c) Repeat MRI in 6 weeks and monitor femoral head edema with repeat MRI as needed; return to sports when edema has resolved.
   d) Monitor with radiographs alone to look for femoral head collapse.

4. What are MRI indications of athletic pubalgia?
   a) Bone marrow edema at the pubis.
   b) Fluid within the pubic symphysis joint.
   c) Widening of the pubic symphysis.
   d) Fluid beneath the pubic bone (within the aponeurosis).
   e) All of the above.

5. What are clinical and imaging findings associated with idiopathic transient osteoporosis? (choose all that apply)
   a) Most commonly in third trimester of pregnancy and in middle-aged men.
   b) Fat-suppressed MRI shows diffuse high-signal-intensity edema in the femoral head and proximal femoral neck and a small hip joint effusion.
   c) MRI shows focal signal changes at the weight-bearing surface of the femoral head.
   d) MRI shows low-signal, crescent-shaped rim in a subchondral location.

References

[1] Weber AE, Poultsides LA, O'Sullivan E, Kelly BT, Bedi A. Anatomical considerations and clinical examination. In: Kelly BT, Bedi A, Larson CM, O'Sullivan E, eds. Sports hip injuries: diagnosis and management. Thorofare, NJ: SLACK Incorporated; 2015:3–24

[2] Kassarjian A, Yoon LS, Belzile E, Connolly SA, Millis MB, Palmer WE. Triad of MR arthrographic findings in patients with cam-type femoroacetabular impingement. Radiology. 2005; 236(2):588–592

[3] Kramer J, Recht MP. MR arthrography of the lower extremity. Radiol Clin North Am. 2002; 40(5):1121–1132

[4] Schmid MR, Nötzli HP, Zanetti M, Wyss TF, Hodler J. Cartilage lesions in the hip: diagnostic effectiveness of MR arthrography. Radiology. 2003; 226(2): 382–386

[5] Potter HG, Foo LF. Magnetic resonance imaging of articular cartilage: trauma, degeneration, and repair. Am J Sports Med. 2006; 34(4):661–677

[6] Mintz DN, Hooper T, Connell D, Buly R, Padgett DE, Potter HG. Magnetic resonance imaging of the hip: detection of labral and chondral abnormalities using noncontrast imaging. Arthroscopy. 2005; 21(4):385–393

[7] Toomayan GA, Holman WR, Major NM, Kozlowicz SM, Vail TP. Sensitivity of MR arthrography in the evaluation of acetabular labral tears. AJR Am J Roentgenol. 2006; 186(2):449–453

[8] Kelly BT, Williams RJ, III, Philippon MJ. Hip arthroscopy: current indications, treatment options, and management issues. Am J Sports Med. 2003; 31(6): 1020–1037

[9] Ferguson SJ, Bryant JT, Ganz R, Ito K. An in vitro investigation of the acetabular labral seal in hip joint mechanics. J Biomech. 2003; 36(2):171–178

[10] McCarthy JC, Noble PC, Schuck MR, Wright J, Lee J. The Otto E. Aufranc Award: the role of labral lesions to development of early degenerative hip disease. Clin Orthop Relat Res. 2001(393):25–37

[11] Dinauer PA, Murphy KP, Carroll JF. Sublabral sulcus at the posteroinferior acetabulum: a potential pitfall in MR arthrography diagnosis of acetabular labral tears. AJR Am J Roentgenol. 2004; 183(6):1745–1753

[12] Alzaharani A, Bali K, Gudena R, et al. The innervation of the human acetabular labrum and hip joint: an anatomic study. BMC Musculoskelet Disord. 2014; 15:41

[13] Kim YT, Azuma H. The nerve endings of the acetabular labrum. Clin Orthop Relat Res. 1995(320):176–181

[14] Kelly BT, Shapiro GS, Digiovanni CW, Buly RL, Potter HG, Hannafin JA. Vascularity of the hip labrum: a cadaveric investigation. Arthroscopy. 2005; 21(1):3–11

[15] Seldes RM, Tan V, Hunt J, Katz M, Winiarsky R, Fitzgerald RH, Jr. Anatomy, histologic features, and vascularity of the adult acetabular labrum. Clin Orthop Relat Res. 2001(382):232–240

[16] Lee AJ, Armour P, Thind D, Coates MH, Kang AC. The prevalence of acetabular labral tears and associated pathology in a young asymptomatic population. Bone Joint J. 2015; 97-B(5):623–627

[17] Register B, Pennock AT, Ho CP, Strickland CD, Lawand A, Philippon MJ. Prevalence of abnormal hip findings in asymptomatic participants: a prospective, blinded study. Am J Sports Med. 2012; 40(12):2720–2724

[18] Jayakar R, Merz A, Plotkin B, Wang D, Seeger L, Hame SL. Magnetic resonance arthrography and the prevalence of acetabular labral tears in patients 50 years of age and older. Skeletal Radiol. 2016; 45(8):1061–1067

[19] Georgiadis AG, Seeley MA, Chauvin NA, Sankar WN. Prevalence of acetabular labral tears in asymptomatic children. J Child Orthop. 2016; 10(2):149–154

[20] Kelly BT, Weiland DE, Schenker ML, Philippon MJ. Arthroscopic labral repair in the hip: surgical technique and review of the literature. Arthroscopy. 2005; 21(12):1496–1504

[21] Lage LA, Patel JV, Villar RN. The acetabular labral tear: an arthroscopic classification. Arthroscopy. 1996; 12(3):269–272

[22] Blankenbaker DG, De Smet AA, Keene JS, Fine JP. Classification and localization of acetabular labral tears. Skeletal Radiol. 2007; 36(5):391–397

[23] Ilizaliturri VM, Jr, Byrd JWT, Sampson TG, et al. A geographic zone method to describe intra-articular pathology in hip arthroscopy: cadaveric study and preliminary report. Arthroscopy. 2008; 24(5):534–539

[24] Byrd JWT. Labral lesions: an elusive source of hip pain case reports and literature review. Arthroscopy. 1996; 12(5):603–612

[25] Türker M, Kılıçoğlu Ö, Göksan B, Bilgiç B. Vascularity and histology of fetal labrum and chondrolabral junction: its relevance to chondrolabral detachment tears. Knee Surg Sports Traumatol Arthrosc. 2012; 20(2):381–386

[26] Kalhor M, Horowitz K, Beck M, Nazparvar B, Ganz R. Vascular supply to the acetabular labrum. J Bone Joint Surg Am. 2010; 92(15):2570–2575

[27] Beck M, Kalhor M, Leunig M, Ganz R. Hip morphology influences the pattern of damage to the acetabular cartilage: femoroacetabular impingement as a cause of early osteoarthritis of the hip. J Bone Joint Surg Br. 2005; 87(7): 1012–1018

[28] Ganz R, Parvizi J, Beck M, Leunig M, Nötzli H, Siebenrock KA. Femoroacetabular impingement: a cause for osteoarthritis of the hip. Clin Orthop Relat Res. 2003(417):112–120

[29] Ito K, Minka MA, II, Leunig M, Werlen S, Ganz R. Femoroacetabular impingement and the cam-effect. A MRI-based quantitative anatomical study of the femoral head-neck offset. J Bone Joint Surg Br. 2001; 83(2):171–176

[30] McGuffin WS, Melkus G, Rakhra KS, Beaulé PE. Is the contralateral hip at risk in patients with unilateral symptomatic cam femoroacetabular impingement? A quantitative T1ρ MRI study. Osteoarthritis Cartilage. 2015; 23(8):1337–1342

[31] Stoller DW, Sampson T, Bredella M. The hip. In: Stoller DW, ed. Magnetic resonance imaging in orthopaedics and sports medicine. Philadelphia, PA: Lippincott Williams & Wilkins; 2007:41–304

[32] Tanzer M, Noiseux N. Osseous abnormalities and early osteoarthritis: the role of hip impingement. Clin Orthop Relat Res. 2004(429):170–177

[33] Leunig M, Casillas MM, Hamlet M, et al. Slipped capital femoral epiphysis: early mechanical damage to the acetabular cartilage by a prominent femoral metaphysis. Acta Orthop Scand. 2000; 71(4):370–375

[34] Agricola R, Heijboer MP, Ginai AZ, et al. A cam deformity is gradually acquired during skeletal maturation in adolescent and young male soccer players: a prospective study with minimum 2-year follow-up. Am J Sports Med. 2014; 42(4):798–806

[35] Johnson AC, Shaman MA, Ryan TG. Femoroacetabular impingement in former high-level youth soccer players. Am J Sports Med. 2012; 40(6):1342–1346

[36] Siebenrock KA, Wahab KHA, Werlen S, Kalhor M, Leunig M, Ganz R. Abnormal extension of the femoral head epiphysis as a cause of cam impingement. Clin Orthop Relat Res. 2004(418):54–60

[37] Pollard TC, Villar RN, Norton MR, et al. Genetic influences in the aetiology of femoroacetabular impingement: a sibling study. J Bone Joint Surg Br. 2010; 92(2):209–216

[38] Frank JM, Harris JD, Erickson BJ, et al. Prevalence of femoroacetabular impingement imaging findings in asymptomatic volunteers: a systematic review. Arthroscopy. 2015; 31(6):1199–1204

[39] Potter HG, Schachar J. High resolution noncontrast MRI of the hip. J Magn Reson Imaging. 2010; 31(2):268–278

[40] Nötzli HP, Wyss TF, Stoecklin CH, Schmid MR, Treiber K, Hodler J. The contour of the femoral head-neck junction as a predictor for the risk of anterior impingement. J Bone Joint Surg Br. 2002; 84(4):556–560

[41] Reynolds D, Lucas J, Klaue K. Retroversion of the acetabulum. A cause of hip pain. J Bone Joint Surg Br. 1999; 81(2):281–288

[42] Bredella MA, Stoller DW. MR imaging of femoroacetabular impingement. Magn Reson Imaging Clin N Am. 2005; 13(4):653–664

[43] Siebenrock KA, Schoeniger R, Ganz R. Anterior femoro-acetabular impingement due to acetabular retroversion. Treatment with periacetabular osteotomy. J Bone Joint Surg Am. 2003; 85(2):278–286

[44] Lavigne M, Parvizi J, Beck M, Siebenrock KA, Ganz R, Leunig M. Anterior femoroacetabular impingement: part I. Techniques of joint preserving surgery. Clin Orthop Relat Res. 2004(418):61–66

[45] Shindle MK, Voos JE, Heyworth BE, et al. Hip arthroscopy in the athletic patient: current techniques and spectrum of disease. J Bone Joint Surg Am. 2007; 89 Suppl 3:29–43

[46] Canham CD, Yen YM, Giordano BD. Does femoroacetabular impingement cause hip instability? A systematic review. Arthroscopy. 2016; 32(1):203–208

[47] Krych AJ, Thompson M, Larson CM, Byrd JW, Kelly BT. Is posterior hip instability associated with cam and pincer deformity? Clin Orthop Relat Res. 2012; 470(12):3390–3397

[48] Magerkurth O, Jacobson JA, Morag Y, Caoili E, Fessell D, Sekiya JK. Capsular laxity of the hip: findings at magnetic resonance arthrography. Arthroscopy. 2013; 29(10):1615–1622

[49] Armstrong DW, III, Rue JPH, Wilckens JH, Frassica FJ. Stress fracture injury in young military men and women. Bone. 2004; 35(3):806–816

[50] Anderson K, Strickland SM, Warren R. Hip and groin injuries in athletes. Am J Sports Med. 2001; 29(4):521–533

[51] Boden BP, Osbahr DC. High-risk stress fractures: evaluation and treatment. J Am Acad Orthop Surg. 2000; 8(6):344–353

[52] Devas MB. Stress fractures of the femoral neck. J Bone Joint Surg Br. 1965; 47 (4):728–738

[53] Egol KA, Koval KJ, Kummer F, Frankel VH. Stress fractures of the femoral neck. Clin Orthop Relat Res. 1998(348):72–78

[54] Fredericson M. Diagnosing tibial stress injuries in athletes. West J Med. 1995; 162(2):150

[55] Blanchard C, Kushare I, Boyles A, Mundy A, Beebe AC, Klingele KE. Traumatic, posterior pediatric hip dislocations with associated posterior labrum osteochondral avulsion: recognizing the acetabular “fleck” sign. J Pediatr Orthop. 2016; 36(6):602–607

[56] Berkes MB, Cross MB, Shindle MK, Bedi A, Kelly BT. Traumatic posterior hip instability and femoroacetabular impingement in athletes. Am J Orthop. 2012; 41(4):166–171

[57] Laorr A, Greenspan A, Anderson MW, Moehring HD, McKinley T. Traumatic hip dislocation: early MRI findings. Skeletal Radiol. 1995; 24(4):239–245

[58] Potter HG, Montgomery KD, Heise CW, Helfet DL. MR imaging of acetabular fractures: value in detecting femoral head injury, intraarticular fragments, and sciatic nerve injury. AJR Am J Roentgenol. 1994; 163(4): 881–886

[59] Moorman CT, III, Warren RF, Hershman EB, et al. Traumatic posterior hip subluxation in American football. J Bone Joint Surg Am. 2003; 85(7):1190–1196

[60] Shindle MK, Ranawat AS, Kelly BT. Diagnosis and management of traumatic and atraumatic hip instability in the athletic patient. Clin Sports Med. 2006; 25(2):309–326, ix–x

[61] Byrd JWT. Lateral impact injury. A source of occult hip pathology. Clin Sports Med. 2001; 20(4):801–815

[62] Palmer WE, Kuong SJ, Elmadbouh HM. MR imaging of myotendinous strain. AJR Am J Roentgenol. 1999; 173(3):703–709

[63] Bencardino JT, Kassarjian A, Palmer WE. Magnetic resonance imaging of the hip: sports-related injuries. Top Magn Reson Imaging. 2003; 14(2): 145–160

[64] Meislin R, Abeles A. Role of hip MR imaging in the management of sports-related injuries. Magn Reson Imaging Clin N Am. 2005; 13(4):635–640

[65] Pavlov H. Roentgen examination of groin and hip pain in the athlete. Clin Sports Med. 1987; 6(4):829–843

[66] Frank JS, Gambacorta PL, Eisner EA. Hip pathology in the adolescent athlete. J Am Acad Orthop Surg. 2013; 21(11):665–674

[67] Zoga AC, Mullens FE, Meyers WC. The spectrum of MR imaging in athletic pubalgia. Radiol Clin North Am. 2010; 48(6):1179–1197

[68] Kunduracioglu B, Yilmaz C, Yorubulut M, Kudas S. Magnetic resonance findings of osteitis pubis. J Magn Reson Imaging. 2007; 25(3):535–539

[69] Khan W, Zoga AC, Meyers WC. Magnetic resonance imaging of athletic pubalgia and the sports hernia: current understanding and practice. Magn Reson Imaging Clin N Am. 2013; 21(1):97–110

[70] Litwin DE, Sneider EB, McEnaney PM, Busconi BD. Athletic pubalgia (sports hernia). Clin Sports Med. 2011; 30(2):417–434

[71] Martin RL, Kivlan BR, Clemente FR. A cadaveric model for ligamentum teres function: a pilot study. Knee Surg Sports Traumatol Arthrosc. 2013; 21(7): 1689–1693

[72] Botser IB, Martin DE, Stout CE, Domb BG. Tears of the ligamentum teres: prevalence in hip arthroscopy using 2 classification systems. Am J Sports Med. 2011; 39 Suppl:117S–125S

[73] Chahla J, Soares EA, Devitt BM, et al. Ligamentum teres tears and femoroacetabular impingement: prevalence and preoperative findings. Arthroscopy. 2016; 32(7):1293–1297

[74] Mayes S, Ferris AR, Smith P, Garnham A, Cook J. Atraumatic tears of the ligamentum teres are more frequent in professional ballet dancers than a sporting population. Skeletal Radiol. 2016; 45(7):959–967

[75] Papavasiliou A, Siatras T, Bintoudi A, et al. The gymnasts' hip and groin: a magnetic resonance imaging study in asymptomatic elite athletes. Skeletal Radiol. 2014; 43(8):1071–1077

[76] Dwek J, Pfirrmann C, Stanley A, Pathria M, Chung CB. MR imaging of the hip abductors: normal anatomy and commonly encountered pathology at the greater trochanter. Magn Reson Imaging Clin N Am. 2005; 13(4):691–704, vii

[77] Chung CB, Robertson JE, Cho GJ, Vaughan LM, Copp SN, Resnick D. Gluteus medius tendon tears and avulsive injuries in elderly women: imaging findings in six patients. AJR Am J Roentgenol. 1999; 173(2):351–353

[78] Kingzett-Taylor A, Tirman PF, Feller J, et al. Tendinosis and tears of gluteus medius and minimus muscles as a cause of hip pain: MR imaging findings. AJR Am J Roentgenol. 1999; 173(4):1123–1126

[79] Bunker TD, Esler CNA, Leach WJ. Rotator-cuff tear of the hip. J Bone Joint Surg Br. 1997; 79(4):618–620

[80] Howell GED, Biggs RE, Bourne RB. Prevalence of abductor mechanism tears of the hips in patients with osteoarthritis. J Arthroplasty. 2001; 16(1):121–123

[81] LaBan MM, Weir SK, Taylor RS. "Bald trochanter" spontaneous rupture of the conjoined tendons of the gluteus medius and minimus presenting as a trochanteric bursitis. Am J Phys Med Rehabil. 2004; 83(10):806–809

[82] Jacobson T, Allen WC. Surgical correction of the snapping iliopsoas tendon. Am J Sports Med. 1990; 18(5):470–474

[83] Hernando MF, Cerezal L, Pérez-Carro L, Abascal F, Canga A. Deep gluteal syndrome: anatomy, imaging, and management of sciatic nerve entrapments in the subgluteal space. Skeletal Radiol. 2015; 44(7):919–934

[84] Papadopoulos EC, Korres DS, Papachristou G, Efstathopoulos N. Piriformis syndrome. Orthopedics. 2004; 27(8):797–799, 799, author reply 799

[85] O'Sullivan E, Antolak S, Martin HD. Nerve compression injuries about the hip and deep gluteal syndrome. In: Kelly BT, Bedi A, Larson CM, O'Sullivan E, eds. Sports hip injuries: diagnosis and management. Thorofare, NJ: SLACK Incorporated; 2015:153–170

[86] Felson DT. An update on the pathogenesis and epidemiology of osteoarthritis. Radiol Clin North Am. 2004; 42(1):1–9, v

[87] Sadro C. Current concepts in magnetic resonance imaging of the adult hip and pelvis. Semin Roentgenol. 2000; 35(3):231–248

[88] Blankenbaker DG, Tuite MJ. MR imaging of early hip joint degeneration. Magn Reson Imaging Clin N Am. 2011; 19(2):365–378

[89] Gold SL, Burge AJ, Potter HG. MRI of hip cartilage: joint morphology, structure, and composition. Clin Orthop Relat Res. 2012; 470(12):3321–3331

[90] Cooperman D. What is the evidence to support acetabular dysplasia as a cause of osteoarthritis? J Pediatr Orthop. 2013; 33 Suppl 1:S2–S7

[91] Klaue K, Durnin CW, Ganz R. The acetabular rim syndrome. A clinical presentation of dysplasia of the hip. J Bone Joint Surg Br. 1991; 73(3): 423–429

[92] Murphy SB, Ganz R, Müller ME. The prognosis in untreated dysplasia of the hip. A study of radiographic factors that predict the outcome. J Bone Joint Surg Am. 1995; 77(7):985–989

[93] Reijman M, Hazes JM, Pols HA, Koes BW, Bierma-Zeinstra SM. Acetabular dysplasia predicts incident osteoarthritis of the hip: the Rotterdam study. Arthritis Rheum. 2005; 52(3):787–793

[94] Fujii M, Nakashima Y, Yamamoto T, et al. Acetabular retroversion in developmental dysplasia of the hip. J Bone Joint Surg Am. 2010; 92(4): 895–903

[95] Kim WY, Hutchinson CE, Andrew JG, Allen PD. The relationship between acetabular retroversion and osteoarthritis of the hip. J Bone Joint Surg Br. 2006; 88(6):727–729

[96] Tönnis D, Heinecke A. Acetabular and femoral anteversion: relationship with osteoarthritis of the hip. J Bone Joint Surg Am. 1999; 81(12):1747–1770

[97] Atweh LA, Kan JH. Multimodality imaging of developmental dysplasia of the hip. Pediatr Radiol. 2013; 43 Suppl 1:S166–S171

[98] Assouline-Dayan Y, Chang C, Greenspan A, Shoenfeld Y, Gershwin ME. Pathogenesis and natural history of osteonecrosis. Semin Arthritis Rheum. 2002; 32(2):94–124

[99] Coleman BG, Kressel HY, Dalinka MK, Scheibler ML, Burk DL, Cohen EK. Radiographically negative avascular necrosis: detection with MR imaging. Radiology. 1988; 168(2):525–528

[100] Hauzeur JP, Pasteels JL, Schoutens A, et al. The diagnostic value of magnetic resonance imaging in non-traumatic osteonecrosis of the femoral head. J Bone Joint Surg Am. 1989; 71(5):641–649

[101] Mitchell DG, Steinberg ME, Dalinka MK, Rao VM, Fallon M, Kressel HY. Magnetic resonance imaging of the ischemic hip. Alterations within the osteonecrotic, viable, and reactive zones. Clin Orthop Relat Res. 1989 (244):60–77

[102] Totty WG, Murphy WA, Ganz WI, Kumar B, Daum WJ, Siegel BA. Magnetic resonance imaging of the normal and ischemic femoral head. AJR Am J Roentgenol. 1984; 143(6):1273–1280

[103] Hayes CW, Balkissoon AA. Magnetic resonance imaging of the musculoskeletal system. II. The hip. Clin Orthop Relat Res. 1996(322):297–309

[104] Shimizu K, Moriya H, Akita T, Sakamoto M, Suguro T. Prediction of collapse with magnetic resonance imaging of avascular necrosis of the femoral head. J Bone Joint Surg Am. 1994; 76(2):215–223

[105] Takatori Y, Kamogawa M, Kokubo T, et al. Magnetic resonance imaging and histopathology in femoral head necrosis. Acta Orthop Scand. 1987; 58(5): 499–503

[106] Khanna AJ, Yoon TR, Mont MA, Hungerford DS, Bluemke DA. Femoral head osteonecrosis: detection and grading by using a rapid MR imaging protocol. Radiology. 2000; 217(1):188–192

[107] Lafforgue P, Dahan E, Chagnaud C, Schiano A, Kasbarian M, Acquaviva PC. Early-stage avascular necrosis of the femoral head: MR imaging for prognosis in 31 cases with at least 2 years of follow-up. Radiology. 1993; 187(1):199–204

[108] Koo KH, Kim R. Quantifying the extent of osteonecrosis of the femoral head. A new method using MRI. J Bone Joint Surg Br. 1995; 77(6):875–880

[109] Cherian SF, Laorr A, Saleh KJ, Kuskowski MA, Bailey RF, Cheng EY. Quantifying the extent of femoral head involvement in osteonecrosis. J Bone Joint Surg Am. 2003; 85(2):309–315

[110] Beltran J, Herman LJ, Burk JM, et al. Femoral head avascular necrosis: MR imaging with clinical-pathologic and radionuclide correlation. Radiology. 1988; 166(1 Pt 1):215–220

[111] Beltran J, Opsha O. MR imaging of the hip: osseous lesions. Magn Reson Imaging Clin N Am. 2005; 13(4):665–676, vi

[112] Kalliakmanis AG, Pneumaticos S, Plessas S, Papachristou G. Transient hip osteoporosis. Orthopedics. 2006; 29(3):263–264

[113] Klontzas ME, Vassalou EE, Zibis AH, Bintoudi AS, Karantanas AH. MR imaging of transient osteoporosis of the hip: an update on 155 hip joints. Eur J Radiol. 2015; 84(3):431–436

[114] Pedersen AB, Mor A, Mehnert F, Thomsen RW, Johnsen SP, Nørgaard M. Rheumatoid arthritis: trends in antirheumatic drug use, C-reactive protein levels, and surgical burden. J Rheumatol. 2015; 42(12):2247–2254

[115] Koulouris G, Morrison WB. MR imaging of hip infection and inflammation. Magn Reson Imaging Clin N Am. 2005; 13(4):743–755

[116] Burki V, Gossec L, Payet J, et al. Prevalence and characteristics of hip involvement in spondyloarthritis: a single-centre observational study of 275 patients. Clin Exp Rheumatol. 2012; 30(4):481–486

[117] Chen HA, Chen CH, Liao HT, et al. Factors associated with radiographic spinal involvement and hip involvement in ankylosing spondylitis. Semin Arthritis Rheum. 2011; 40(6):552–558

[118] Vander Cruyssen B, Muñoz-Gomariz E, Font P, et al. ASPECT-REGISPONSER-RESPONDIA working group. Hip involvement in ankylosing spondylitis: epidemiology and risk factors associated with hip replacement surgery. Rheumatology (Oxford). 2010; 49(1):73–81

[119] Lee S, Lee JY, Hwang JH, Shin JH, Kim TH, Kim SK. Clinical importance of inflammatory facet joints of the spine in ankylosing spondylitis: a magnetic resonance imaging study. Scand J Rheumatol. 2016; 45(6):491–498

[120] Szypryt P, Twining P, Preston BJ, Howell CJ. Synovial chondromatosis of the hip joint presenting as a pathological fracture. Br J Radiol. 1986; 59(700): 399–401

[121] Ferro FP, Philippon MJ. Arthroscopy provides symptom relief and good functional outcomes in patients with hip synovial chondromatosis. J Hip Preserv Surg. 2015; 2(3):265–271

[122] Tyler WK, Vidal AF, Williams RJ, Healey JH. Pigmented villonodular synovitis. J Am Acad Orthop Surg. 2006; 14(6):376–385

[123] Dietrich TJ, Dora C, Pfirrmann CW. Postoperative imaging in femoroacetabular impingement. Semin Musculoskelet Radiol. 2013; 17(3):272–278

[124] Clohisy JC, Nepple JJ, Larson CM, Zaltz I, Millis M, Academic Network of Conservation Hip Outcome Research (ANCHOR) Members. Persistent structural disease is the most common cause of repeat hip preservation surgery. Clin Orthop Relat Res. 2013; 471(12):3788–3794

[125] Zingg PO, Buehler TC, Poutawera VR, Alireza A, Dora C. Femoral neck fractures after arthroscopic femoral neck osteochondroplasty for femoroacetabular impingement. Knee Surg Sports Traumatol Arthrosc. 2014; 22(4):926–931

[126] Beck M. Groin pain after open FAI surgery: the role of intraarticular adhesions. Clin Orthop Relat Res. 2009; 467(3):769–774

[127] Blankenbaker DG, De Smet AA, Keene JS. MR arthrographic appearance of the postoperative acetabular labrum in patients with suspected recurrent labral tears. AJR Am J Roentgenol. 2011; 197(6):W1118–22

[128] Bedi A, Zbeda RM, Bueno VF, Downie B, Dolan M, Kelly BT. The incidence of heterotopic ossification after hip arthroscopy. Am J Sports Med. 2012; 40(4): 854–863

Answers to Common Clinical Questions

1. Answer: b. Coronal imaging provides visualization of the superior labrum, but sagittal cuts best show the superior/anterior labrum, where most labral tears occur. Although MR arthrography is still used, we prefer T1 noncontrast imaging for the best visualization of labral pathology.
2. Answer: b: The prevalence of asymptomatic CAM lesions is 37%, asymptomatic pincer lesions is 67%, and asymptomatic labral tears in young adults is 33%–69%.
3. Answer: c. Initial MRI may not show evidence of subluxation, and the process may continue to evolve. Although there may be alternative approaches to treatment of hip subluxation, it is the authors' preference to repeat the MRI to monitor for femoral head edema resolution while keeping the patient non-weight-bearing and allowing return to sports when MRI shows resolution of edema.
4. Answer: e. Athletic pubalgia can manifest on MRI as involvement of the insertion of the rectus abdominis, the origin of the adductor longus, or the periosteum of the pubic tubercle.
5. Answer: a and b. These are clinical and MRI findings associated with idiopathic transient osteoporosis. c and d are associated with osteonecrosis of the femoral head.